

ROBERTO NOGUEIRA

AI-Native Engineer | Ruby on Rails | React | Agentic Systems

Software & Telecom Engineer | AI-Native Product Archtiect

São José dos Campos, SP, Brazil



CONTACT

+55 (12) 98124-5001

enogrob@gmail.com

linkedin.com/in/enogrob

enogrob.github.io

enogrob.github.io/projects/

github.com/enogrob

PROFESSIONAL SUMMARY

AI-Native Software **Engineer** with a strong Telecom background and **7+ years** of **Ruby on Rails** experience. Skilled in **Ruby, React, JavaScript, Python,** and **Elixir,** building agentic systems, AI-native knowledge systems, modular monoliths, microservices, scalable backend architectures, CI/CD, automation, and workflow integration. Deep experience in **OSS/BSS,** multinational system integration, **billing, payroll, subscriptions, API initegrations,** mentoring, technical leadership, and reliable software delivery at scale.

VALUE HIGHLIGHTS

- **7+ years** of experience with **Ruby on Rails** and strong backend / full-stack delivery.
- **International engineering experience** across LATAM, USA, Canada, and Europe.
- **300%** productivity improvement through Ruby/Rake automation at Ericsson.
- Experience with **payroll, billing, subscriptions,** event-driven integrations, and async processing.
- **AI-assisted SDLC, agentic workflows, multi-agent systems,** and context engineering.

PORTFOLIO

REAL PROJECTS. REAL IMPACT.



FEATURED PROJECT

HOLOFLUX

A human-centered semantic runtime and visual language for intent, inquiry, meaning, evidence, execution, and evolving understanding.

github.com/enogrob/project-holoflux

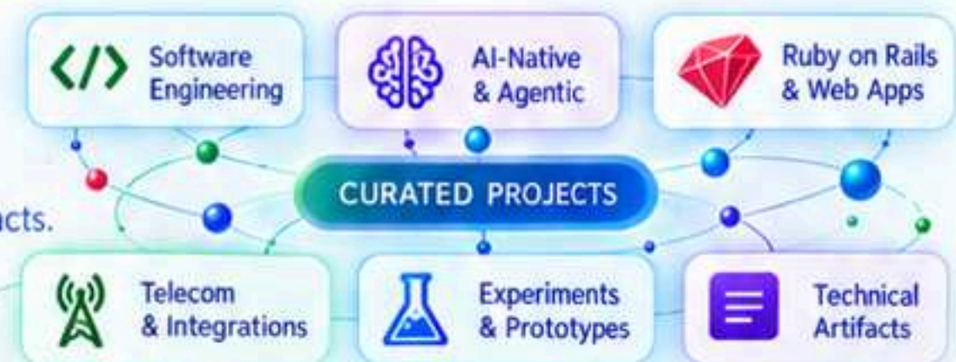




PROJECTS PORTFOLIO

A curated portfolio of software engineering, AI-native, Ruby on Rails, telecom, and product projects spanning platforms, integrations, experiments, and technical artifacts.

enogrob.github.io/projects/



CORE COMPETENCIES

Languages:	Ruby, Python, JavaScript, Elixir, Erlang/OTP, Java
Frameworks & Frontend:	Ruby on Rails, React, Phoenix, Vue, jQuery, Bootstrap
AI & Agentic Engineering:	AI-Native Software Engineering, AI-Assisted SDLC, Agentic Workflows, Multi-Agent Systems, Context Engineering, GitHub Copilot, OpenAI Codex, Claude Code, Warp
Architecture:	Modular Monoliths, Packwerk, Microservices, REST, GraphQL, Clean Architecture, DDD, Event-Driven Architecture, Kafka
Backend & Async Processing:	Sidekiq, Redis, Background Jobs, API Integration
Databases:	PostgreSQL, MySQL, Oracle/PLSQL, MongoDB, Neo4j, Sybase
Testing & Quality:	RSpec, Minitest, PyTest, JUnit, ExUnit, SimpleCov, RubyCritic, Skunk
Security:	Brakeman, bundler-audit, bundler-leak
DevOps & CI/CD:	Git, GitHub, GitHub Actions, GitLab, Bitbucket, Docker, Kubernetes, Terraform, Helm, Puppet
Cloud & Platforms:	AWS, Heroku, Engine Yard, OpenStack, StarlingX, IBM Cloud
Observability & Tools:	Datadog, Kibana, Redash, Rollbar, OpenAPI/Swagger, Postman
Telecom & OSS/BSS:	Ericsson OSS-RC, BSS, MSS, APG, AXE, CPP, MOShell
Engineering Practices:	Clean Code, SOLID, DRY, Design Patterns, Scrum

TURNING COMPLEXITY INTO UNDERSTANDING

PROFESSIONAL EXPERIENCE

1

06/2013 — Present

Zoatworks — São José dos Campos, Brazil

Founder & Consultant | AI-Native Software Engineer

Summary: Providing software engineering and technology consulting focused on AI-native development, Ruby on Rails, systems integration, telecom, and backend platforms.

Key achievements:

- Built multi-agent and AI-first frameworks for eBooks, career workflows,
- technical blogs, and automation.
- Built an event-driven Ruby on Rails microservice with Kafka to orchestrate student subscription workflows and external API integrations.
- Integrated AI-driven workflows into GitHub-based engineering processes.
- Used AI and code-quality tooling to automate Rails technical-debt analysis and modernization prioritization.
- Developed scalable backend services with Ruby, Elixir, and Erlang/OTP.

Projects: AI-native frameworks — COSMOS / HOLOFLUX / IOP / FACTORIES | Anubis/CX (Qeevo) | PicPay (IEBT) | Track Server (ITER) | Adaive (Lab360) | Taxweb Auditor (GSW) | OSS Support (Entel)

2

encora

06/2021 — 09/2024

Encora — Campinas, Brazil

Senior Ruby on Rails Engineer | System Specialist

Summary: Worked on payroll, billing, and Employer Risk systems at Justworks, plus cloud/edge initiatives at Wind River.

Key achievements:

- Developed scalable payroll and billing workflows using Ruby on Rails, React, GraphQL, Redis, and Sidekiq.
- Worked on Workers' Comp billing including rates, policies, eligibility, error handling, fallback, and reconciliation.
- Built reliable asynchronous processing solutions with Sidekiq/Redis, including bulk S3 workflows.
- Maintained a Rackwerk modular monolith and modernized legacy interfaces with React.
- Used CI/CD and observability with GitHub Actions, Datadog, Kibana, Redash, and Rollbar.
- Technical leadership: interviewing, mentoring, and collaboration.

Projects: PEO / Payroll Platform | Time Tracker | WRCP (Wind River)

3

InMov

07/2018 — 06/2021

InMov — São Paulo, Brazil

Senior Ruby on Rails Engineer (Full Stack)

Key achievements:

- Developed Smart City applications using Ruby on Rails, JavaScript, jQuery, and AJAX.
- Led Ruby on Rails upgrades from version 4.2 to 6.0, improving performance, security, and maintainability.
- Created Site Rails Manager, a Bash/Docker CLI toolkit to automate and standardize Ruby on Rails development environments and workflows.

Projects: City Hall Permit System | Site Rails Manager

4

ERICSSON

12/2004 — 06/2013

Ericsson — Indaiatuba, SP, Brazil

Software Developer | Solution Integrator | Project Manager

Key achievements:

- Developed, integrated, and deployed telecom systems in large-scale OSS/BSS environments.
- Led international OSS/BSS initiatives across LATAM, USA, Canada, and Europe.
- Built MOP Automation using Ruby and Rake, achieving 300% productivity improvement.
- Delivered technical training and knowledge transfer in international projects.

Projects: BSS Dual Stack | Full IP Rehoming | MOP Auternation | Revenue Assurance (Vivo) | OSS/BSS Upgrades & Support

EDUCATION

- **1998** — Data Communications — USP
- **1994** — Software Engineering (Postgraduate) — UNIAP
- **1985** — Telecommunications / Electronics Engineering (Bachelor's Degree) — INATEL

CERTIFICATIONS

- SAP Integration — Thomson Reuters
- Solution Integrator Experienced (Job Stage 5) — Ericsson
- Getting and Cleaning Data — Johns Hopkins University
- Functional Programming in Erlang — Kent University
- R Programming — Johns Hopkins University

COURSES & SPECIALIZATIONS

- **2026** — Claude Code: The Complete Course — Udemy
- **2026** — GitHub Copilot CLI — GitHub
- **2025** — AI Automation — Udemy
- **2025** — MCP Crash Course — Udemy
- **2025** — Working with Vue 3 and Go — Udemy
- **2025** — Working with Microservices in Go — Udemy
- **2025** — GitHub Copilot — Udemy
- **2023** — Metaverse — Coursera
- **2019** — Elixir & OTP — The Pragmatic Studio
- **2019** — Elixir: Scalable and Efficient Application Development — Udemy
- **2017** — Specialization in Erlang Programming — Kent University / FutureLearn
- **2016** — Specialization in Ruby on Rails Web Development — Johns Hopkins University / Coursera
- **2015** — Web Applications Architectures — University of New Mexico / Coursera
- **2014** — Specialization in Data Science — Johns Hopkins University / Coursera
- **2010** — Ruby on Rails Advanced — Caelum
- **2010** — Ruby on Rails — Caelum

LANGUAGES

- Portuguese — Native / Bilingual
- English — Full Professional
- Spanish — Intermediate